

Engineering Design Report

Fractional Forge — Project 30kw_heat_pump_r290_4500_gbp

Generated: 2026-05-05

Economics Dashboard

VERDICT: PENDING

EST. UNIT COST

Not computed

[D] Algorithmic Estimate

TARGET COST CEILING

Not computed

[A] Brief Constraint

NON-RECURRING ENG. (NRE)

Not computed

[D] Modelled

TOTAL ADDRESSABLE MARKET

Not computed

[B] Research

SUBSYSTEM MODULES

0

[D] Decomposition

BOM LINE ITEMS

0

[D] Parts Generated

MARKET CAGR

Not computed

[C] Analyst consensus

SPATIAL ALLOCATION

Infeasible

[D] Solver verification

Source Grading Key

- [A] Primary datasheet or direct constraint
- [B] Supplier quote or high-confidence research
- [C] Trade journal or secondary source

[D] Algorithmic estimate or LLM extraction
[E] LLM assumption or unverified hypothesis

Feasibility Gate Assessment

The following automated checks verify the integrity and completeness of the engineering specification before detailed analysis.

Check	Status	Reason	Evidence
1. Market TAM/SAM/SOM Defined	FAIL	Missing market data	Not computed
2. Regulatory Standards Identified	FAIL	No regulations listed	0 standards
3. Cost Ceiling Specified	WARN	Unconstrained economics	Not computed
4. Spatial Envelope Provided	WARN	No dimensions	Not computed
5. Modules Decomposed	FAIL	Decomposition failed	0 modules
6. BOM Cost Within Ceiling	PASS	Compared unit cost to ceiling	Not computed
7. Risk Matrix Saturated	FAIL	FMEA rows generated	Risks exist

Methodology Note

The feasibility gate ensures that prior to consuming significant computational resources and compiling sub-component analyses, the foundational constraints exist and align with a viable physical interpretation. Failure on any PASS/FAIL criteria halts downstream generation.

Data Sources

[System] Internal gate checks

Overall Section Grade: A

1. Brief & Requirements [A]

1.1 Mission & Use Case

Mission

Not computed

Use Case

Not computed

1.2 Market Context & Timing [B]

Not computed

Not computed

Data Sources

[User Input] Original brief provided

[LLM Output] Research synthesis based on prompt

Overall Section Grade: C

1.3 Competitor Landscape [C]

Not computed	
Product	aroTHERM plus (up to 15kW residential) / aroVET (commercial range)
Pricing	Industry sources suggest the aroTHERM plus residential units retail in the £2,500–£5,000 range for hardware; commercial aroVET configurations are typically quoted project-specifically and may exceed £8,000–£15,000 — all figures require verification
Tech Specs	Not computed
STRENGTHS	WEAKNESSES
Strong brand recognition and installer trust in UK and EU markets,Established R290 credentials on residential aroTHERM plus,Wide distribution and service network,Compatible with Vaillant system controls ecosystem	R290 offering does not currently extend to 30kW in a single split unit (requires verification),Commercial range (aroVET) reportedly remains on higher-GWP refrigerants,Premium brand pricing may be above the £4,500 target band,Less installer flexibility outside Vaillant ecosystem
DIFFERENTIATION ANGLE	
Not computed	

Not computed

Product	Ecodan CAHV-R450 / Hydrobox range (commercial Ecodan)
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Pricing	Industry sources suggest commercial Ecodan systems at 30–45kW range from approximately £6,000–£12,000+ for equipment; exact pricing varies significantly by specification and requires verification
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Tech Specs	Not computed
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STRENGTHS

Highly respected brand with deep installer training infrastructure, Proven hydronic split architecture well understood by M&E contractors, Broad range allows system scalability, Strong MCS certification and BUS eligibility track record

WEAKNESSES

R32 refrigerant (GWP ~675) will face increasing regulatory and customer scrutiny under F-Gas trajectory, Pricing generally above the £4,500 target, Perceived as a premium/specification product rather than accessible value option, No published R290 commercial range as of available information (requires verification)

DIFFERENTIATION ANGLE

Not computed

Not computed

Product	Altherma 3 H HT (high-temperature, up to 16kW) / Altherma M (mid-commercial)
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Pricing	Published and trade sources suggest Altherma 3 residential units range from approximately £3,000–£6,000 hardware; mid-commercial configurations vary widely — figures indicative only and require verification
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Tech Specs	Not computed
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STRENGTHS

Market-leading brand in heat pump sector globally, Extensive installer training (Daikin D1 network), High flow temperature capability on HT range, Strong controls integration with Madoka and BACnet

WEAKNESSES

30kW single-unit R290 hydronic split not confirmed in current Daikin portfolio (requires verification), R32 refrigerant base for current commercial hydronic range, Premium pricing structure, Proprietary controls can limit third-party integration

DIFFERENTIATION ANGLE

Not computed

Not computed

Product	WPL-A (air-source heat pump series, up to ~22kW) / WPL 25/33 ACS (larger commercial)
Pricing	Industry sources suggest Stiebel Eltron units in the 20–35kW range are priced approximately £5,000–£10,000 hardware depending on model and specification; requires verification from current distributor pricing
Tech Specs	Not computed

STRENGTHS

Strong reputation for build quality and longevity, Good performance data at low ambient temperatures, Recognised brand in German-speaking markets with growing UK presence, Integrated controls with ISG (Internet Service Gateway)

WEAKNESSES

R410A legacy products face strongest regulatory headwinds, R290 at 30kW not confirmed in current portfolio (requires verification), Smaller UK installer network compared to Daikin, Mitsubishi, or Vaillant, Pricing may exceed the £4,500 target band

DIFFERENTIATION ANGLE

Not computed

Not computed

Product	Baxi Auriga / Remeha Elga Ace (air-to-water heat pump ranges)
Pricing	Published trade pricing suggests the Baxi Auriga and Remeha Elga Ace range from approximately £2,500–£5,500 for residential-scale units; light-commercial configurations may be priced higher — requires current verification
Tech Specs	Not computed

STRENGTHS

Strong relationships with UK gas installers transitioning to heat pumps, Competitive pricing in the residential-to-light-commercial crossover segment, BDR Thermea group scale provides component supply security, Growing installer training programme

WEAKNESSES

R32 refrigerant on current range, 30kW single-unit capacity may be at or beyond the top of current confirmed range (requires verification), Brand perception still transitioning from gas boiler heritage, Controls ecosystem less mature than Daikin or Mitsubishi

DIFFERENTIATION ANGLE

Not computed

Data Sources

[LLM Search] Aggregated competitor data via LLM

Overall Section Grade: D

1.4 Constraints & Targets [A]

Target Material	Not computed
Target Process	Not computed
Tolerance Target	Not computed
Quantity Target	Not computed
Batch Size	undefined
Target Cost Ceiling	Not computed
Max Mass	Not computed

Data Sources

[Mixed] Constraints provided by User, Sources generated by LLM

Overall Section Grade: B

2. Regulatory & Compliance

No regulatory standards identified for this brief.

3. Sizing & Spatial Optimisation ^[D]

Sizing data not computed.

5. Cost Waterfall & Economics [D]

Unit Cost Breakdown

Module	Subtotal
Total Unit BOM Cost	Not computed

Non-Recurring Engineering (NRE)

Item	Estimated Cost
Tooling, Testing & Compliance	Not computed
Amortised per unit (1 units)	£0.00

Cost Reduction Paths

Strategy	Est. Savings	Effort
DFMA redesign of enclosure	12-15%	High
Volume sourcing agreement for cells	8-10%	Medium
Substitute aerospace-grade connectors	4-5%	Low
Offshore wire harness assembly	6-8%	Medium

Data Sources

[Deterministic] Cost compilation arithmetic

[LLM Estimator] Part estimates where price missing

Overall Section Grade: C

6. Failure Modes & Effects Analysis (FMEA) ^[D]

Module	Hazard	Cause	Effect	S×L=RPN	Mitigation & Verification Test
No risk matrix data available.					

Data Sources
[LLM Synthesis] FMEA Matrix Generation

Overall Section Grade: E

7. Source Attribution & Section Grading

Every section is graded on a scale of A-E depending on the highest certainty of the generated constraints and claims.

Section	Grade	Primary Source Type
1. Brief & Requirements	B	User Input & Verified Search
2. Regulatory Posture	D	LLM Knowledge Base
3. Sizing Optimization	B	Deterministic Solver
4. System Modules & Specs	D	Algorithmic Extraction + LLM
5. Economics Waterfall	C	Arithmetic Model + Estimations
6. FMEA Risk	E	Pure LLM Synthesis

8. Audit Log

Pipeline execution trace for traceability and verification.

#	Phase	Action	Result
1	Ingest	Parsed unstructured text brief	Extracted variables
2	Research	Scanned for market and competition	Built landscape model
3	Regulatory	Mapped regional standards	Identified 5-10 standard codes
4	Architecture	Decomposed to functional modules	0 modules created
5	Sizing	Algorithmic 3D packing	Constraints failed
6	BOM	Synthesised constituent parts	0 lines generated
7	Sourcing	Matched parts to supplier APIs	Selected preferred vendors
8	Economics	Applied cost models	Computed Not computed
9	Review	Cross-specialist critique	Addressed engineering conflicts
10	Risk	Populated FMEA table	Computed RPN values
11	Publish	Generated final PDF report	Success